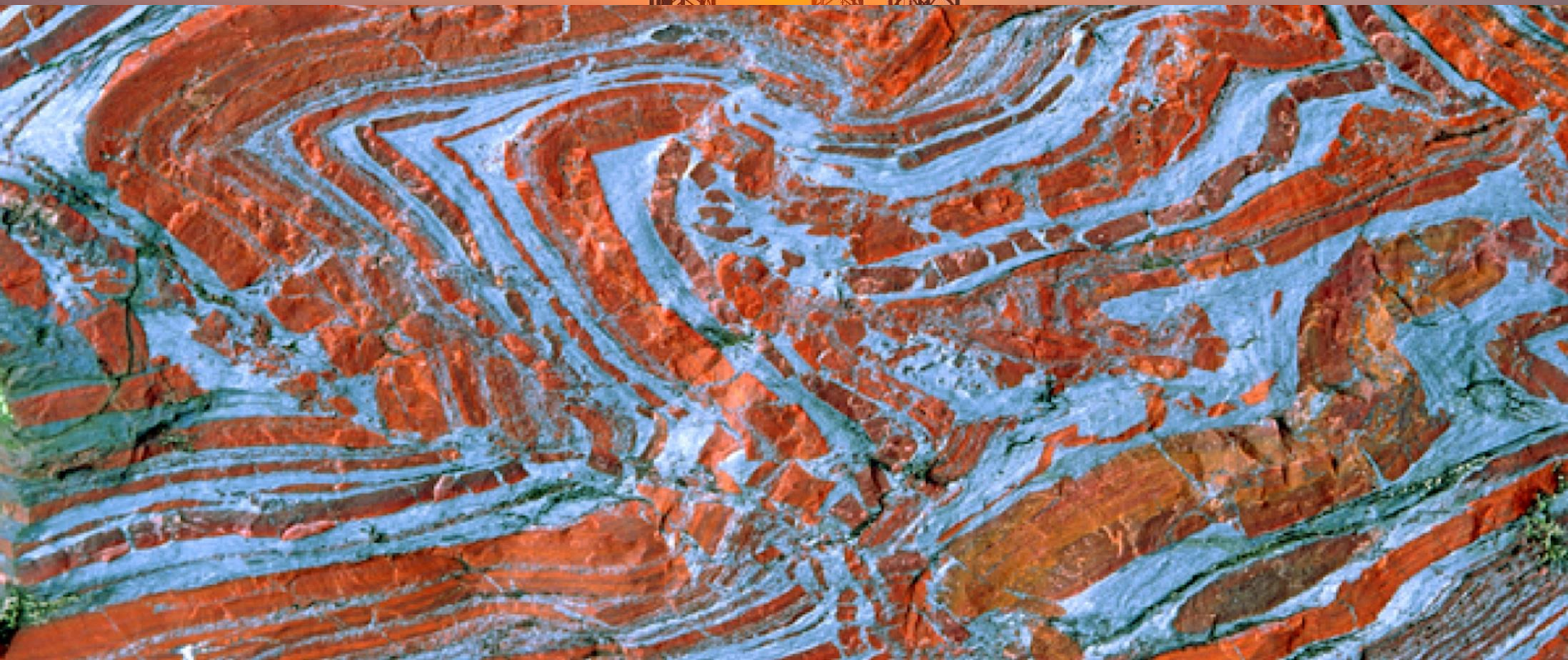
EART 26001
Energy and Resources

Dr Stefan Schroeder

Lecture 11: Global geological trends for commodity deposits

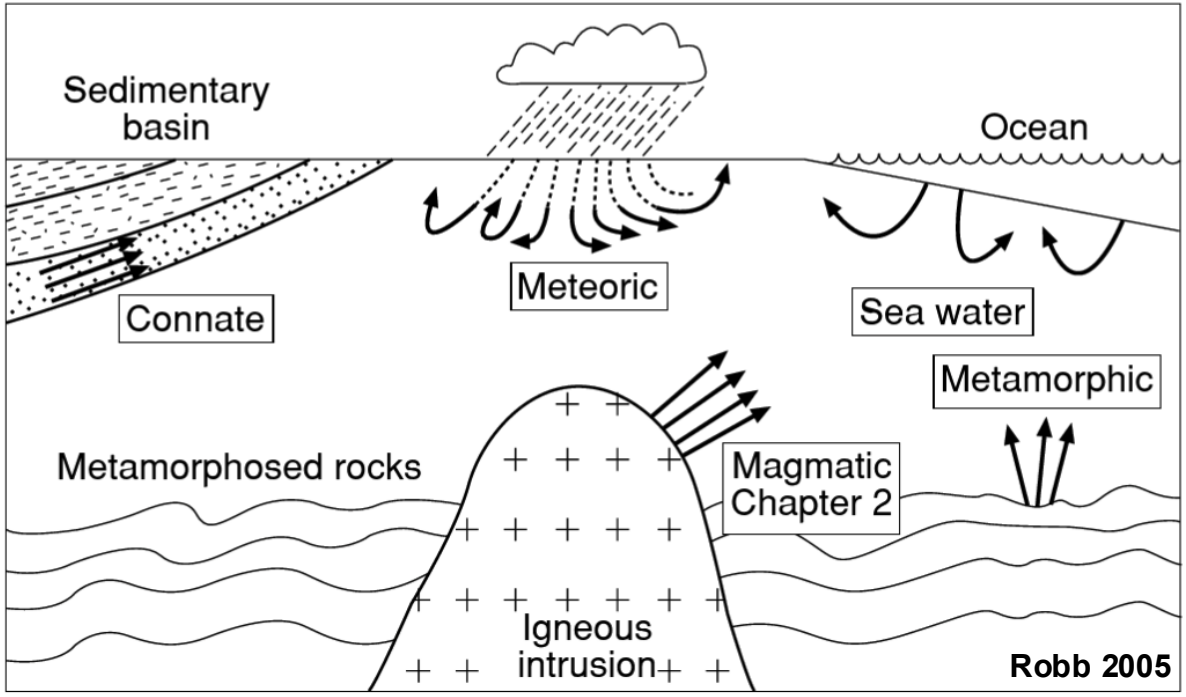


Global geological trends for commodity deposits
Learning goals week 11

This week you will learn to:

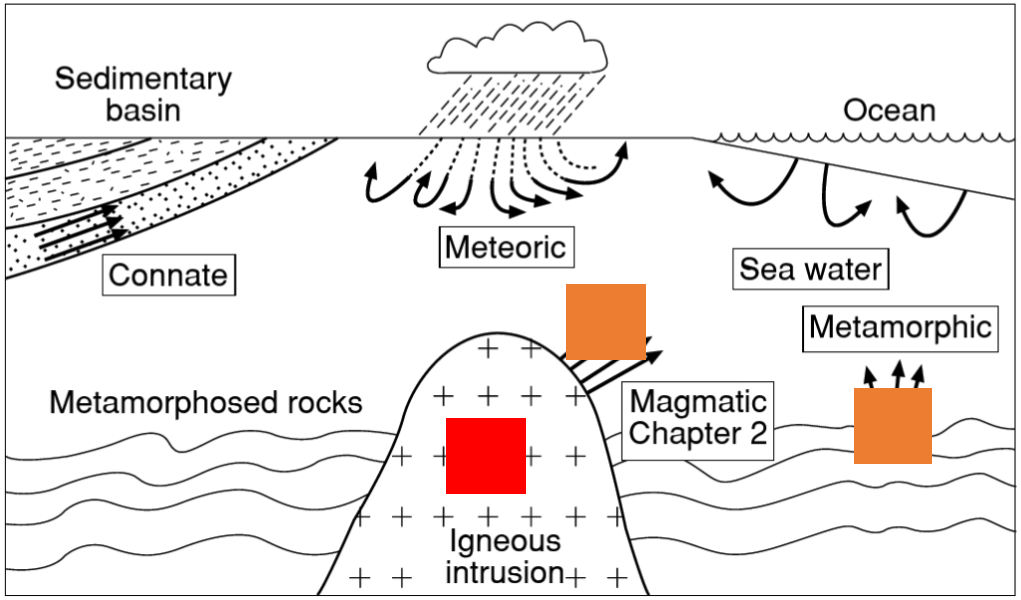
- **Summarize a simple classification scheme of resource deposits**
- **Describe how temporal changes in global geological processes influence commodity deposits (atmospheric, biological, eustatic and geodynamic trends).**

"Resource deposits form when a useful commodity is sufficiently concentrated in an accessible part of the crust so it can be profitably extracted." (Robb 2005)



Classification of resource deposits

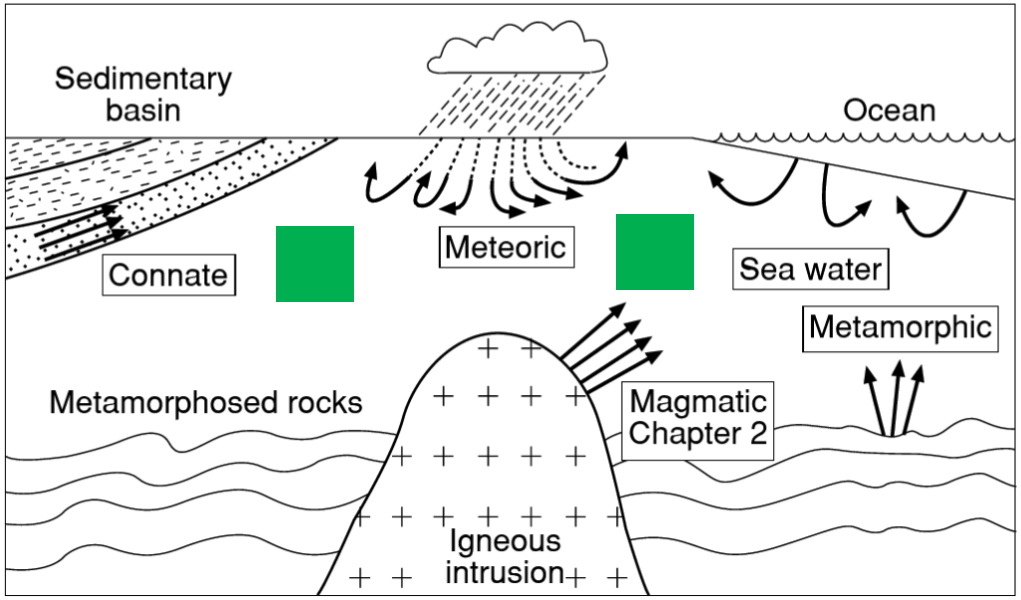
Deposit group	Examples
Igneous/Magmatic • Syngenetic igneous processes associated with magma formation, cooling, crystallization	PGEs, Cr in layered intrusions; diamondiferous kimberlites; lithophile and incompatible elements (U, Li, Sn, W….) in igneous intrusions
Magmatic-hydrothermal • Epigenetic precipitation from aqueous fluids derived from melt	Orogenic Au Other deposit types: • Sn-W fluid flow around granites (Cornwall) – greisen • Porphyry Cu-Mo • Volcanic-epithermal Au • Sediment alteration by igneous intrusion – W-Cu-Zn-Mo skarn



Robb 2005

Classification of resource deposits

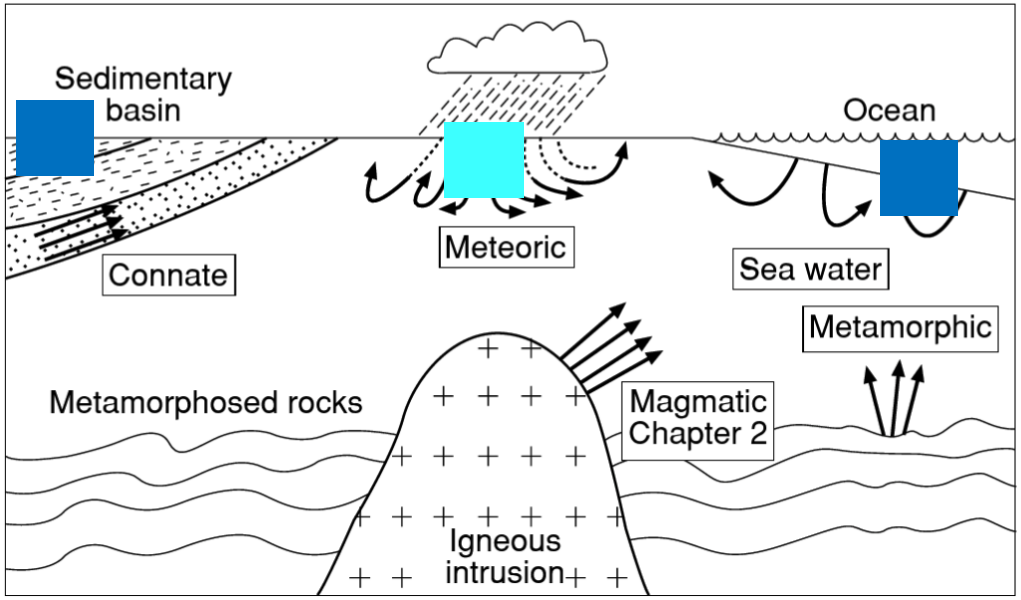
Deposit group	Examples
Hydrothermal • Epigenetic precipitation from hydrothermal aqueous fluids of variable (igneous, metamorphic, connate, surface) origin	• Hydrothermal U (incl sandstone, unconformity deposits); orogenic Au • Other deposit types: • Pb-Zn mineralization in sediments – MVT • Polymetallic exhalative venting and black smokers on seafloor – VMS/Sedex • Stratiform Cu • Epigenetic Au



Robb 2005

Classification of resource deposits

Deposit group	Examples
Sedimentary • Syngenetic processes during sediment formation	Sedimentary Fe-Mn (BIF/MnF); Hydrocarbons; Evaporites; Placer (Au, U, diamonds, Ti, Zr….)
Surficial and supergene • Epigenetic formation from surface fluids • Surface alteration of existing deposits	Supergene Fe; Al residual laterite deposits (bauxite) Other deposit types: • Supergene Cu • Laterite residual Ni and PGEs • Clay deposits (kaolinite)

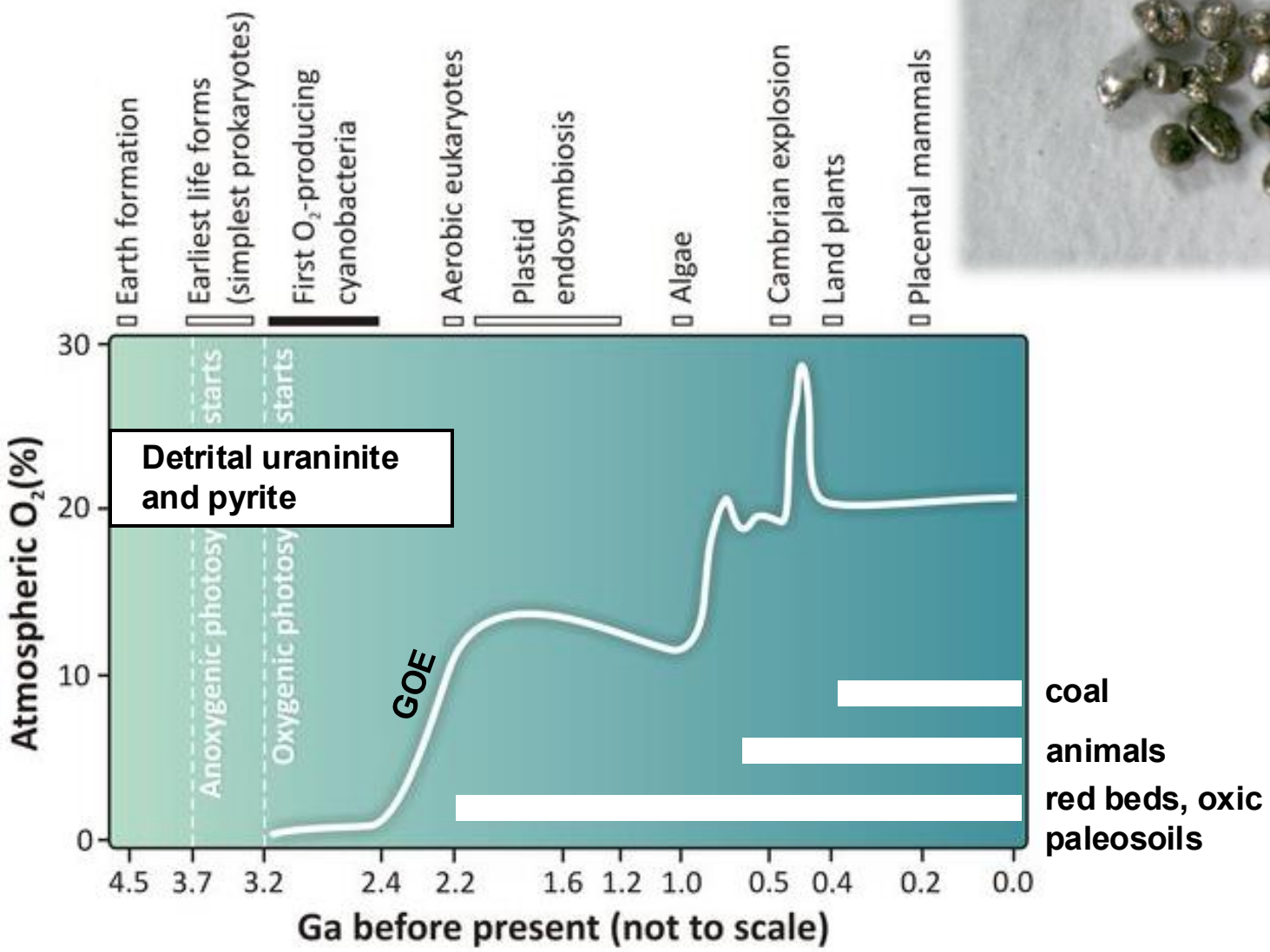


Robb 2005


Atmospheric trends

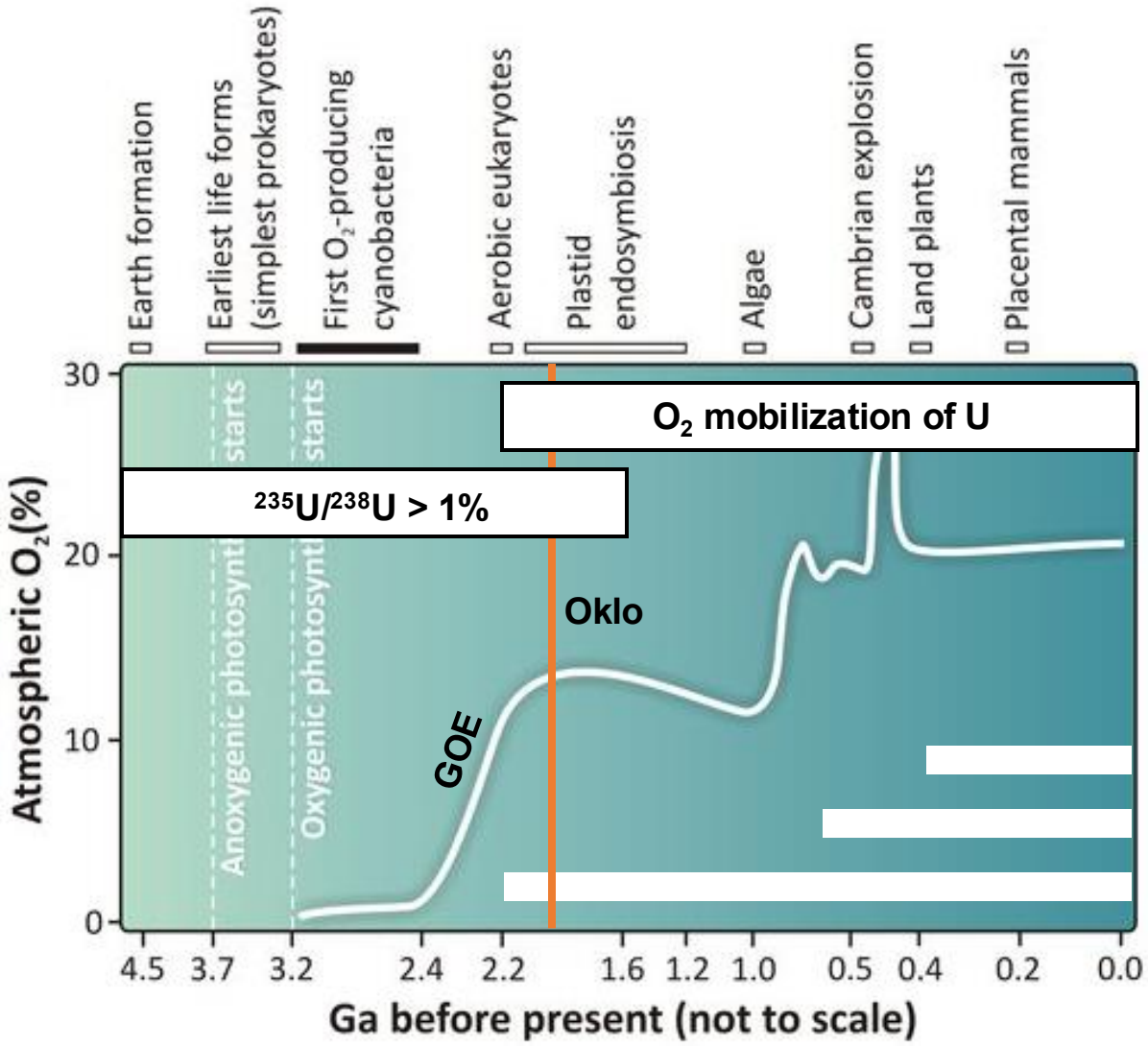
Atmospheric trends (oxygen)
Preservation of redox-sensitive minerals





© NASA Astrobiology

Atmospheric trends (oxygen)
Preservation of redox-sensitive minerals



coal

animals

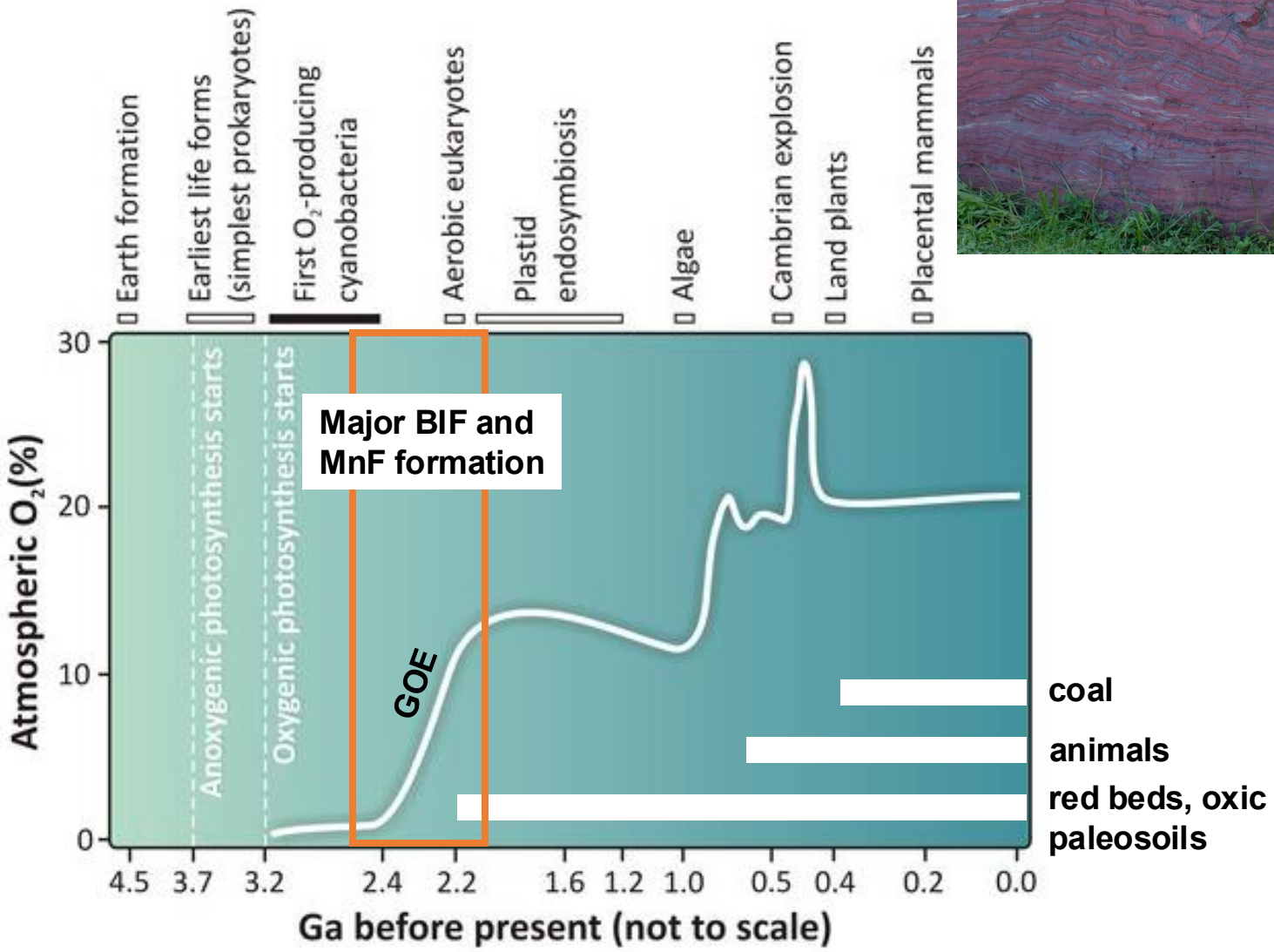
red beds, oxic paleosoils



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Atmospheric trends (oxygen)
Accumulation of redox-sensitive Fe and Mn



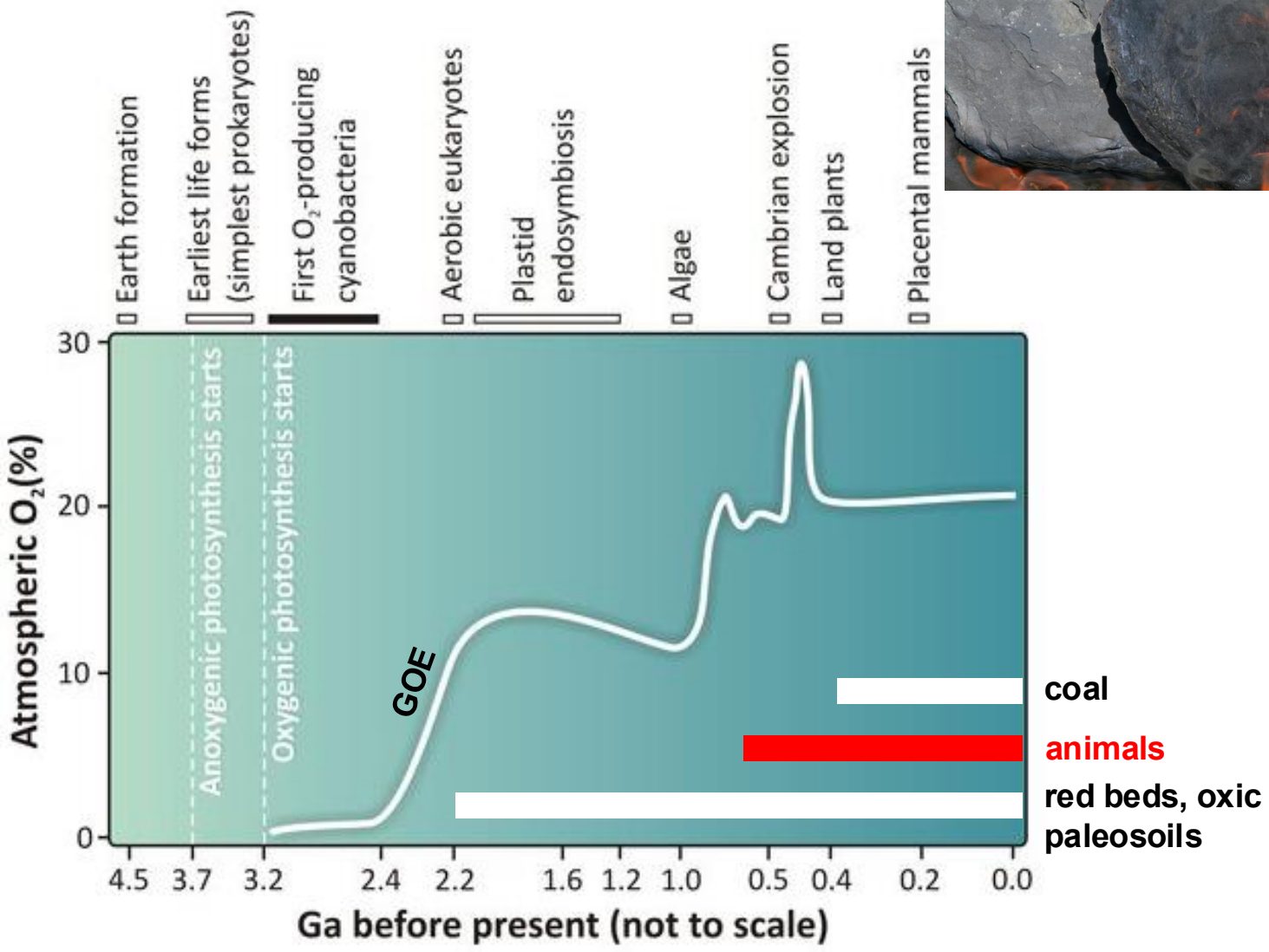


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Biological trends
500 µm

Biological trends
Animal evolution and source rocks





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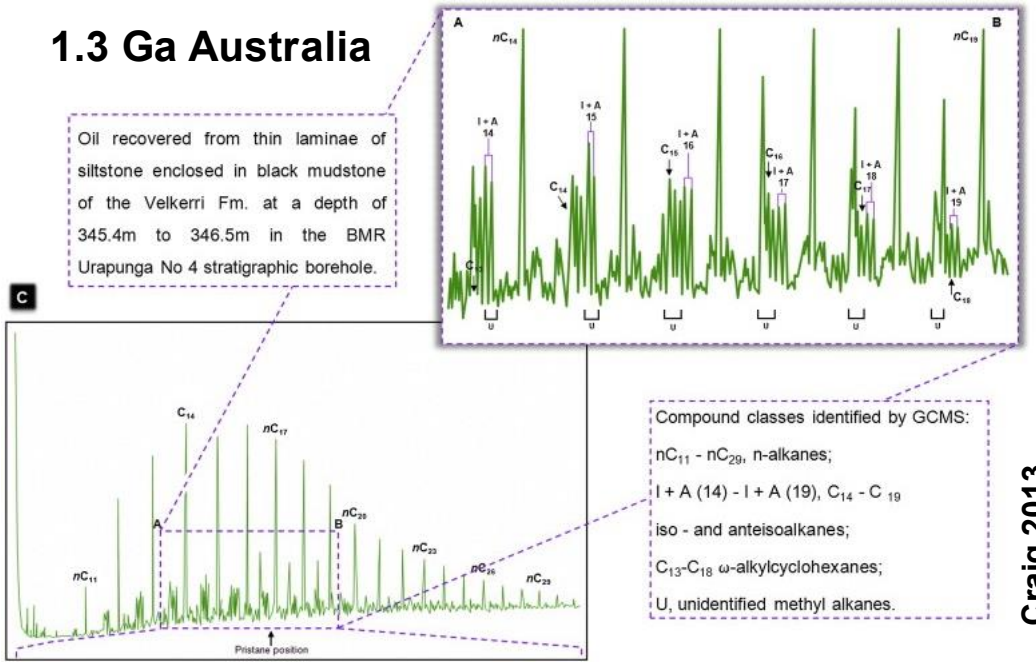
Biological trends
Evolution and source rocks

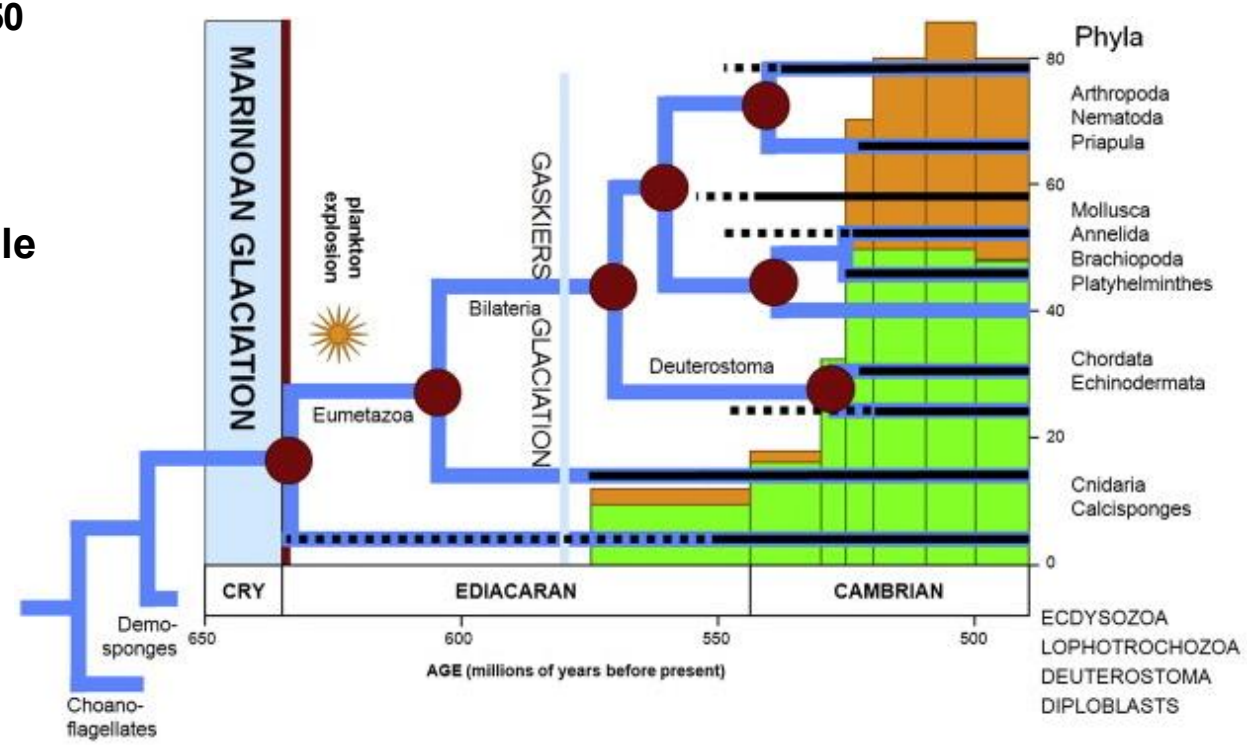


Craig 2013

Biological trends
Evolution and source rocks

- Earliest kerogens dominated by Archaea and microbial mats
- More complex kerogens in later Proterozoic due to:
  - Multicellular eukaryotes ~1.3 Ga
  - Heterotrophic protists ~850 Ma
  - Metazoa ~650 Ma
- Neoproterozoic glacials likely played important role in diversification





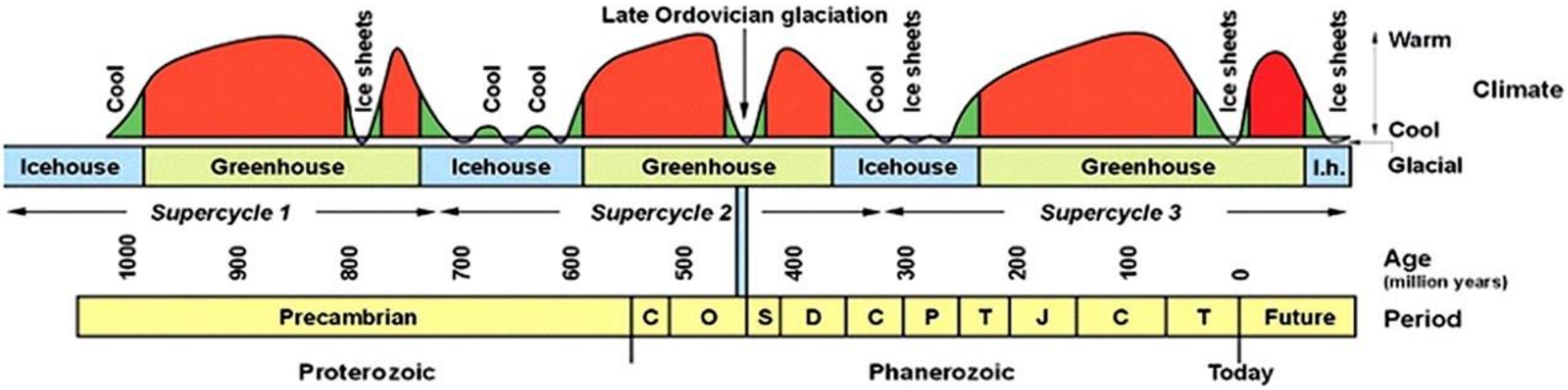

Eustatic trends

Eustatic trends
Sea level and climate

- Sea level is influenced by
  - Plate tectonics – controlling basin volume
  - Glacial (polar) ice volumes
- 'greenhouse' climate: warm periods without polar ice
- 'icehouse climate: cool periods with polar ice

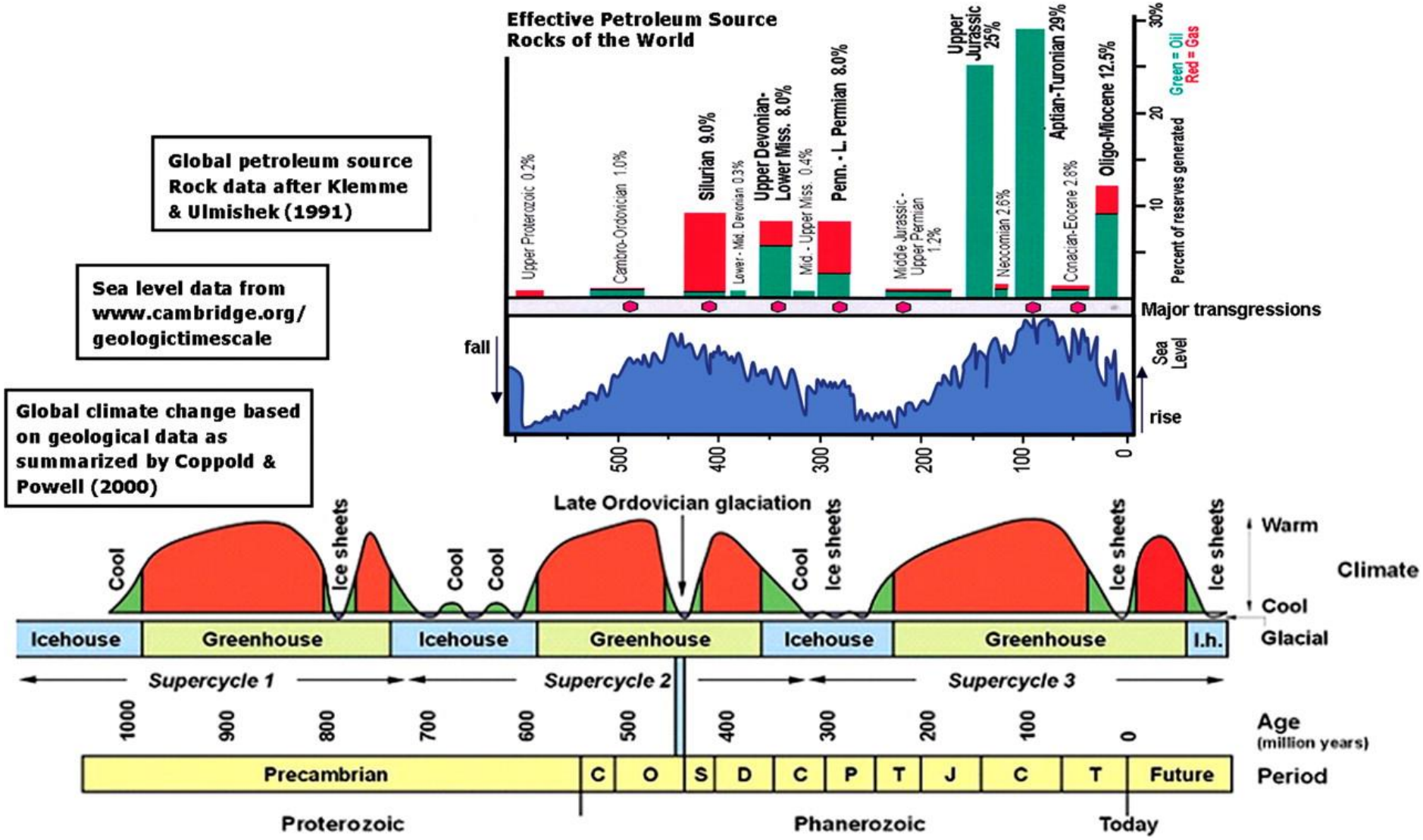


http://neatorama.cachefly.net/images/uploads/2007/01/manhattan_global_warmingjpg.jpg



Craig 2009

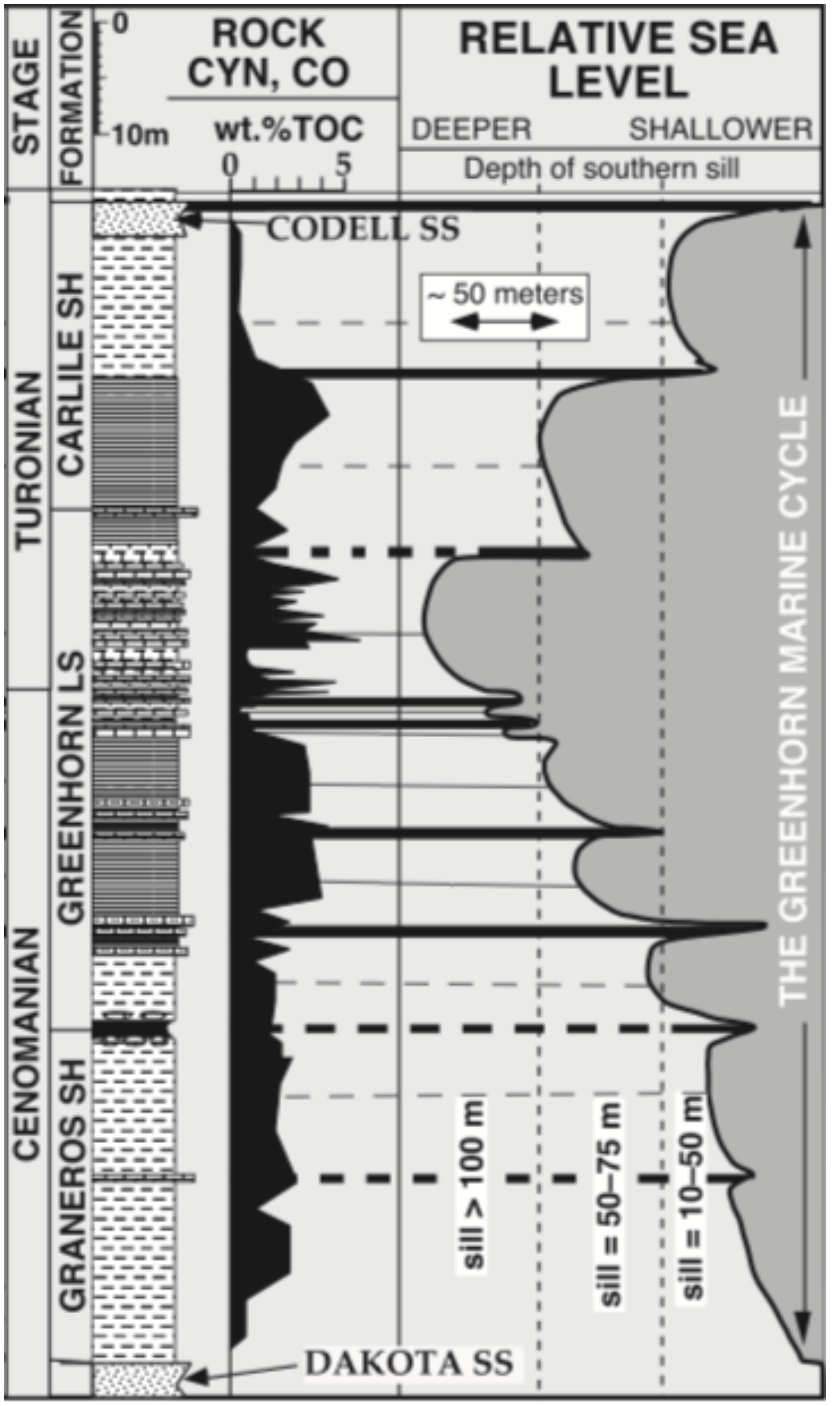
Eustatic trends
Sea level and source rocks



Craig 2009

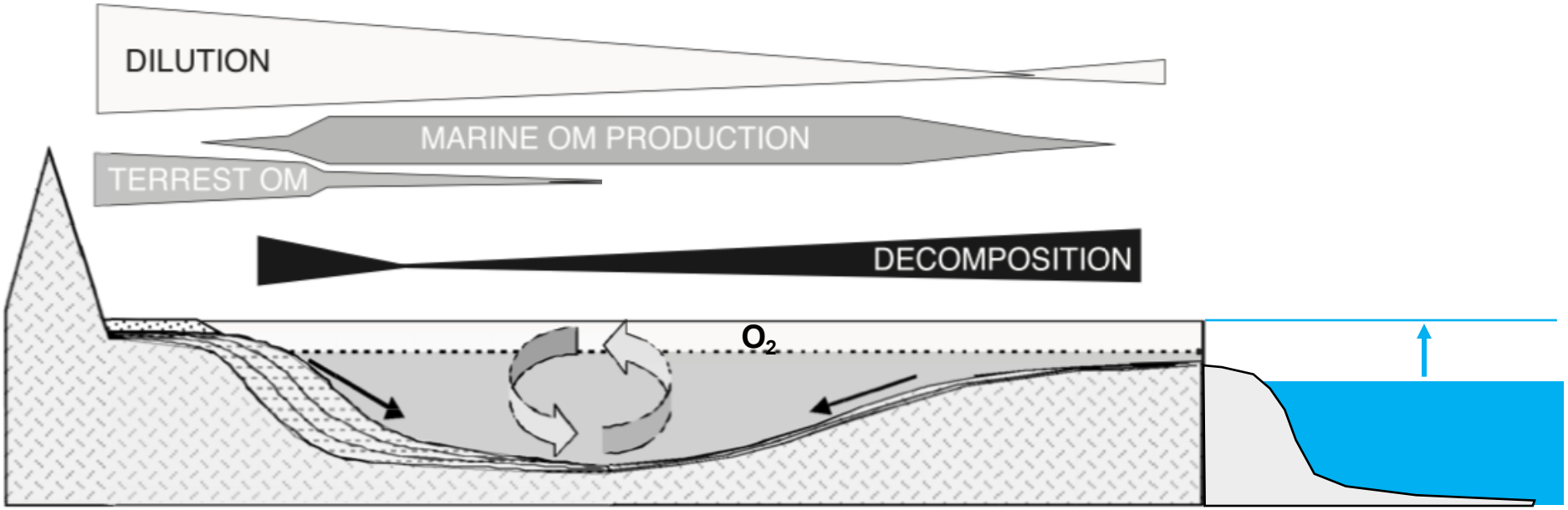
Eustatic trends
Sea level and source rocks





Arthur 2004

Eustatic trends
Sea level and source rocks



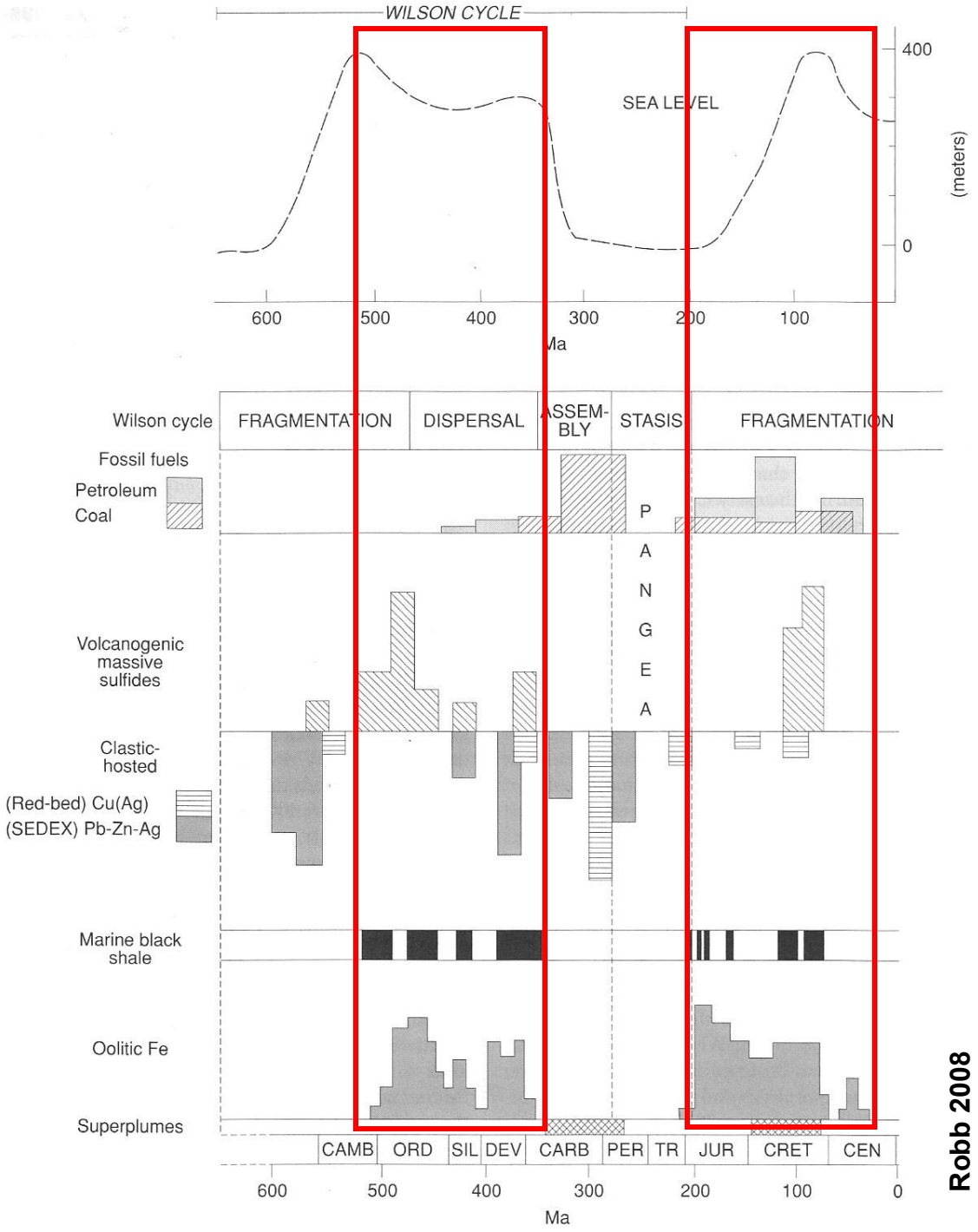
Arthur 2004

Transgressions:

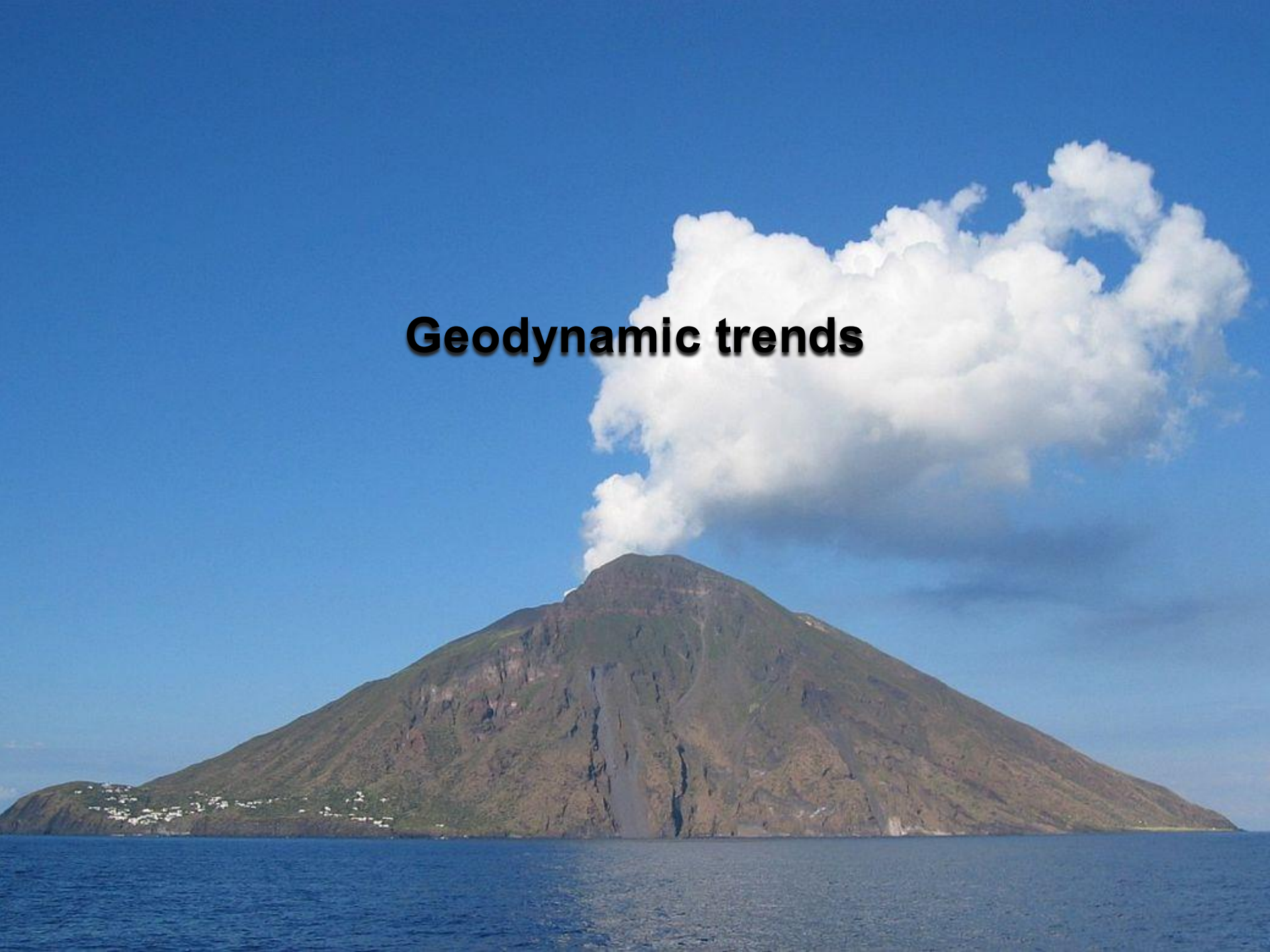
- Advect deeper-water nutrients to shelf: increased productivity
- Cause retention of nutrients in shallow seas: increased productivity
- Flood shallow shelves: increased productivity
- Promote stratification and dys/anoxia: increased preservation
- Cause retention of clastics nearshore: less dilution

Eustatic trends
Sedimentary resources

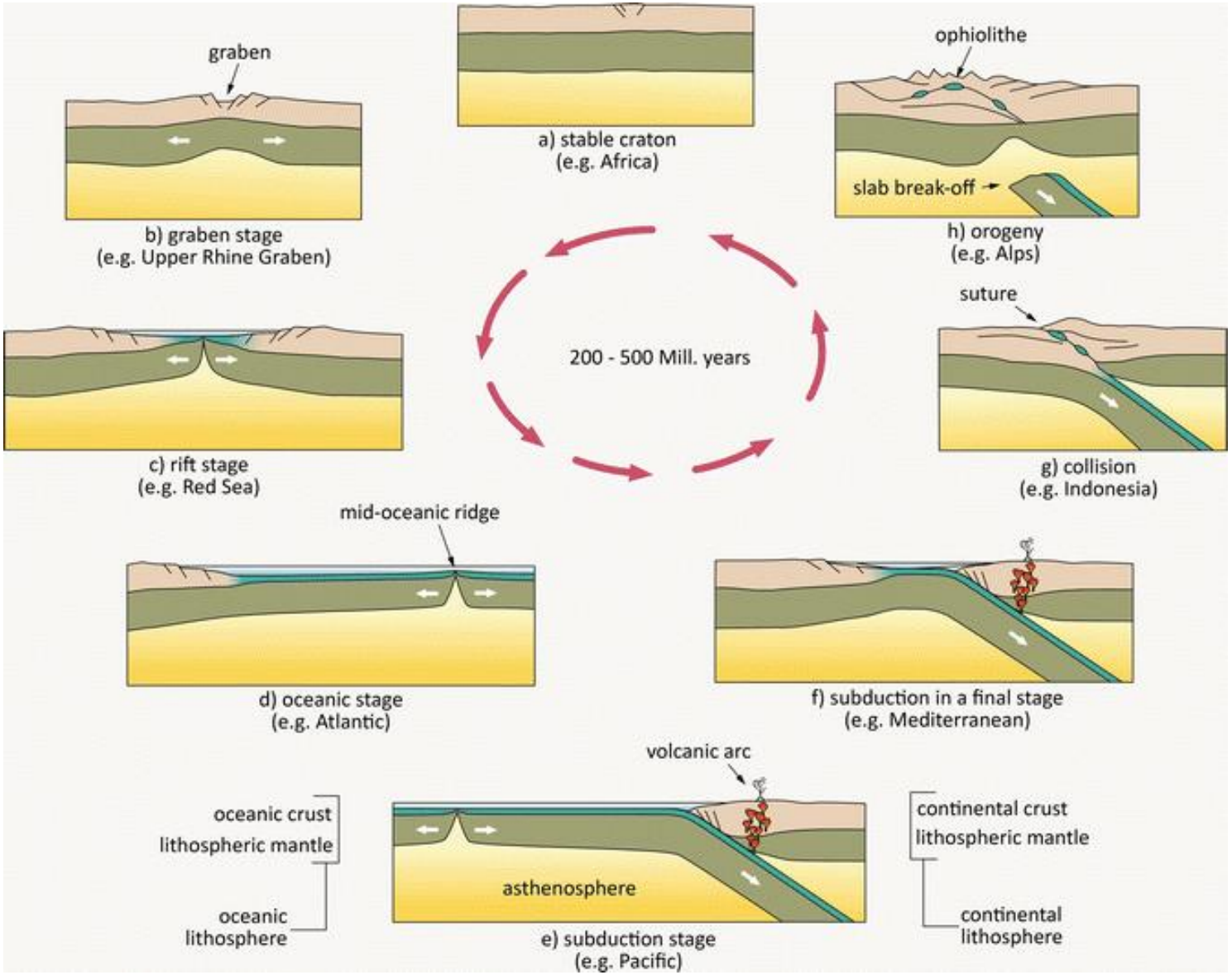
- **Transgression preserves shelfal deposits: marine placer, BIF and oolitic ironstones….**
- **Regression and continental exposure destroy existing deposits**



Robb 2008

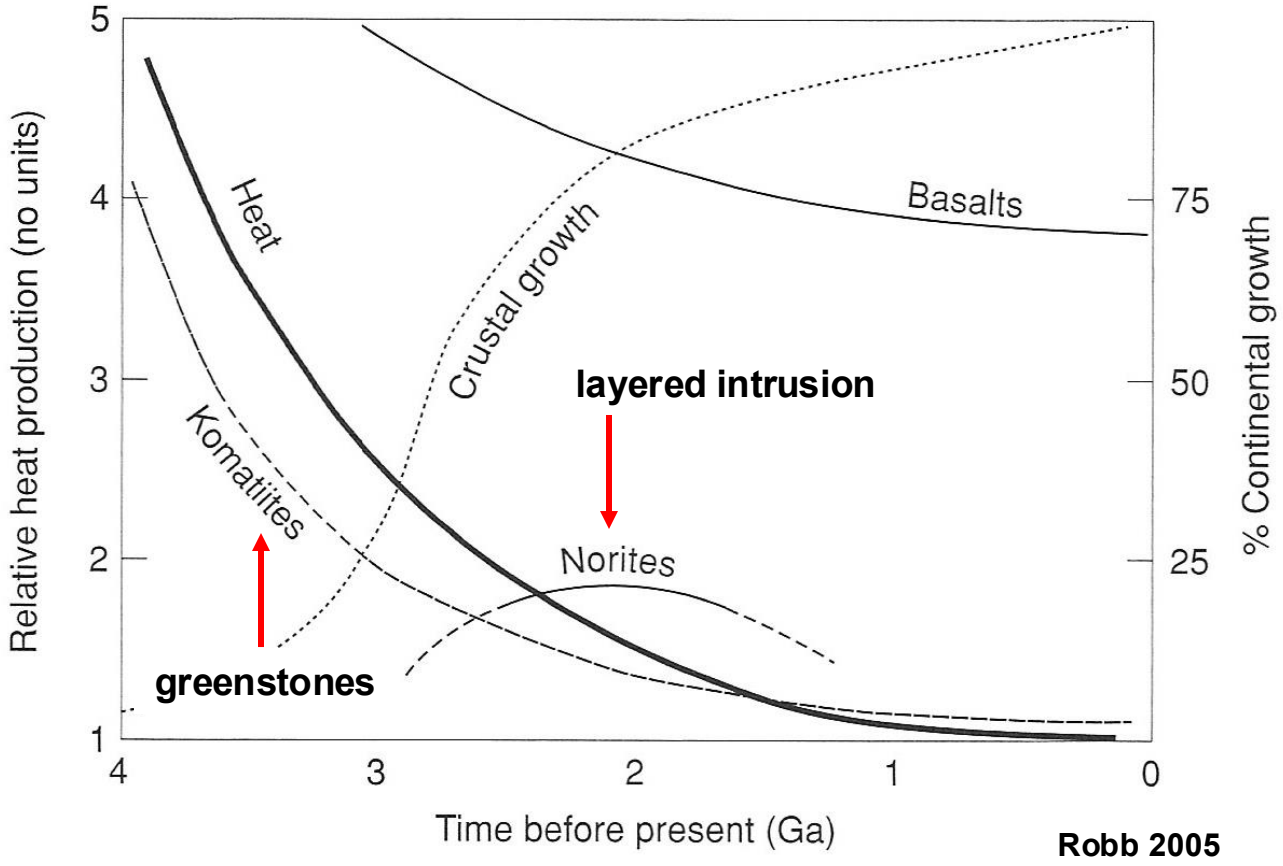

Geodynamic trends

Geodynamic trends
Wilson cycle



Meschede 2016

Geodynamic trends
Heat production



Robb 2005

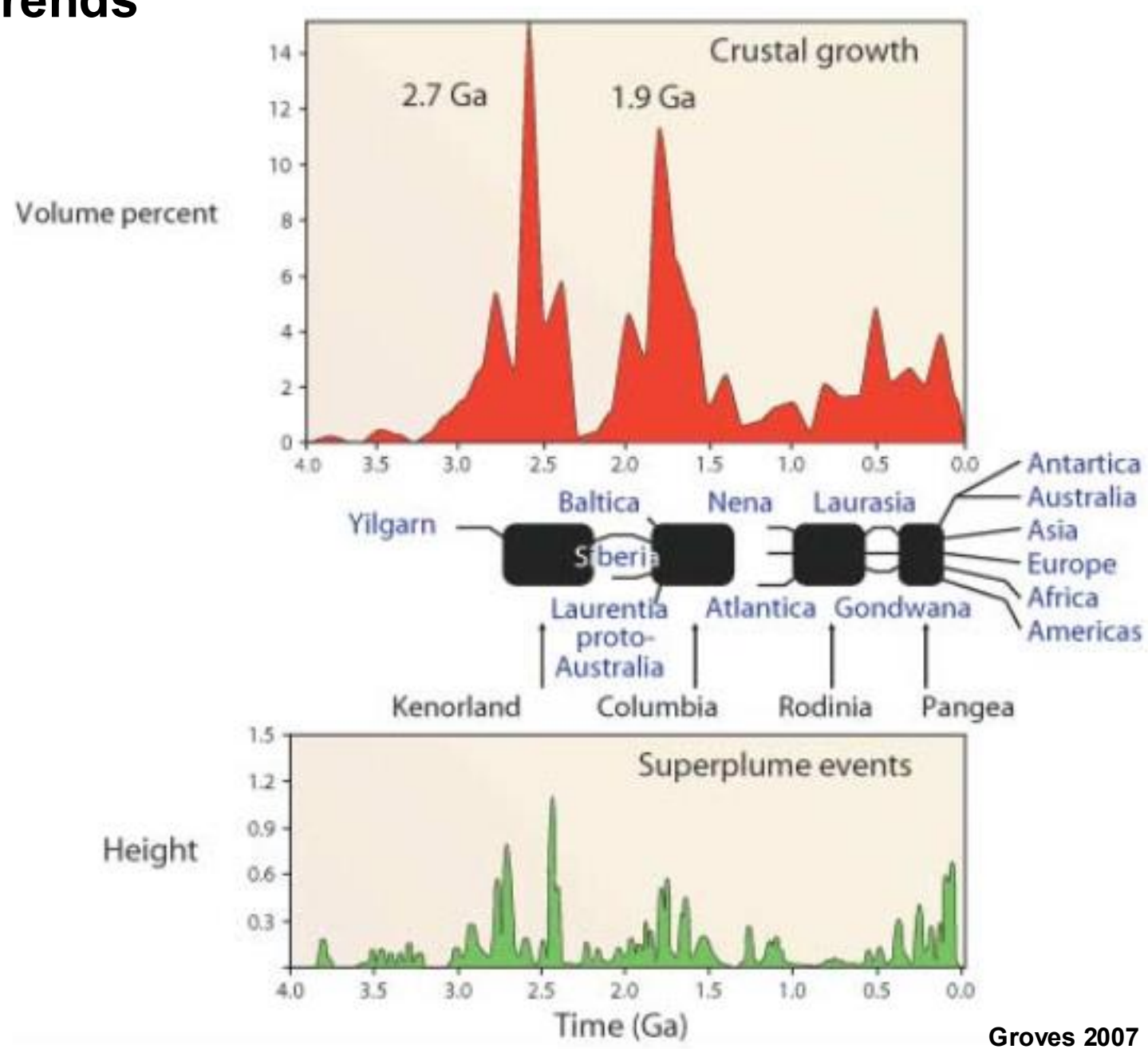
Increased Archean crustal heat production and plume activity:

- **Partial melting and transfer of compatible elements from mantle to newly formed crust: fertile greenstone belts - Au**

Phase of crustal growth at Archean/Proterozoic transition:

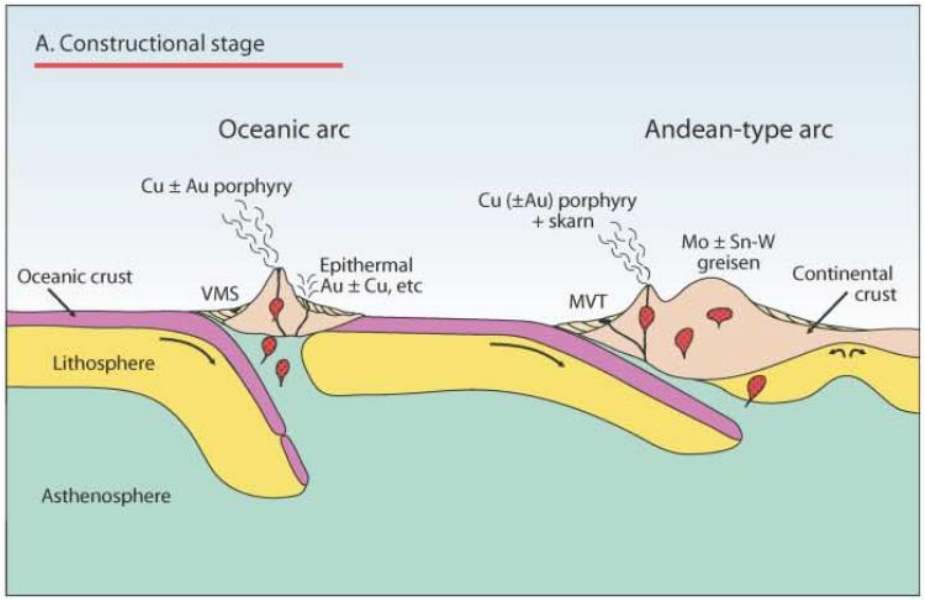
- **Partial melting in lithospheric mantle to produce ultramafic intrusions: Bushveld**

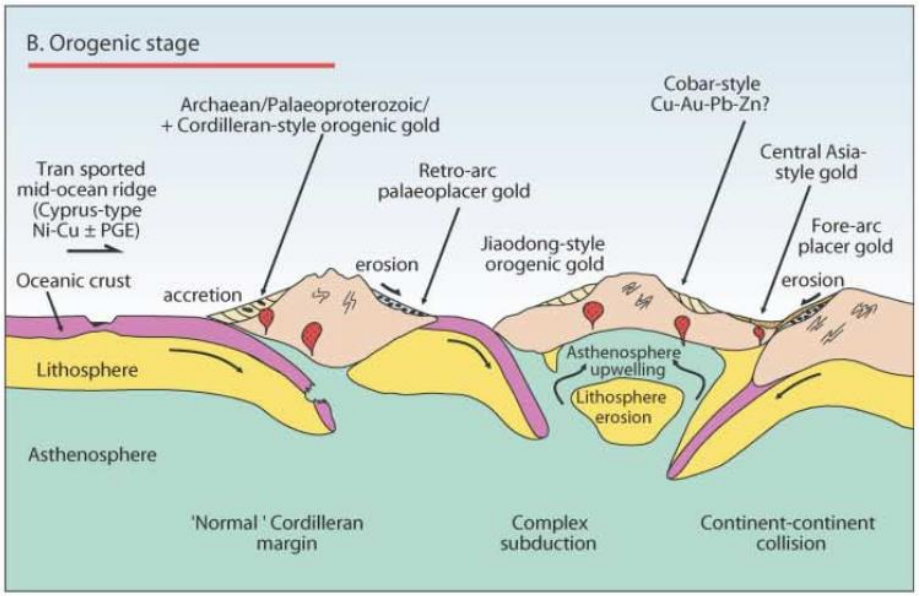
Geodynamic trends
Crustal growth



Groves 2007

Geodynamic trends
Convergent margin/subduction zone



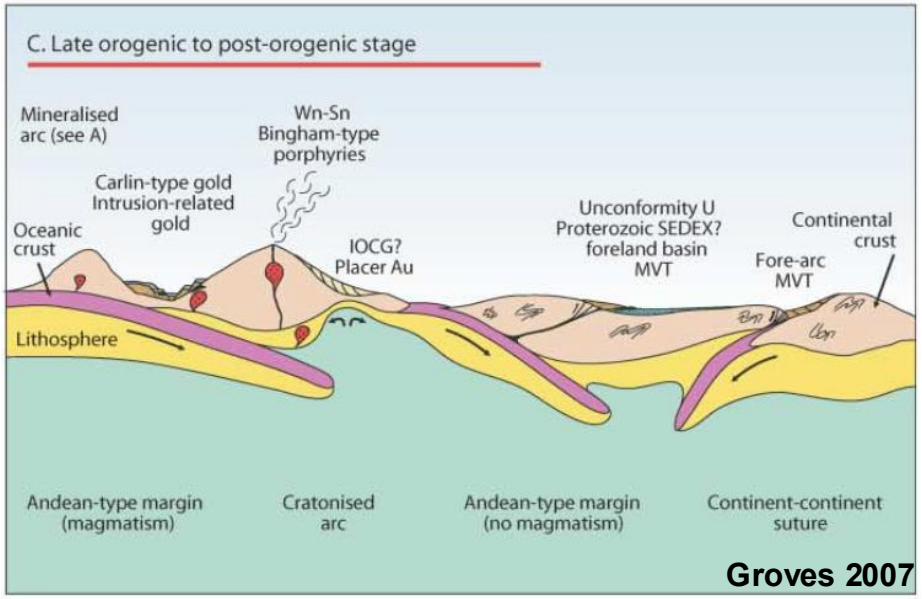




© USGS

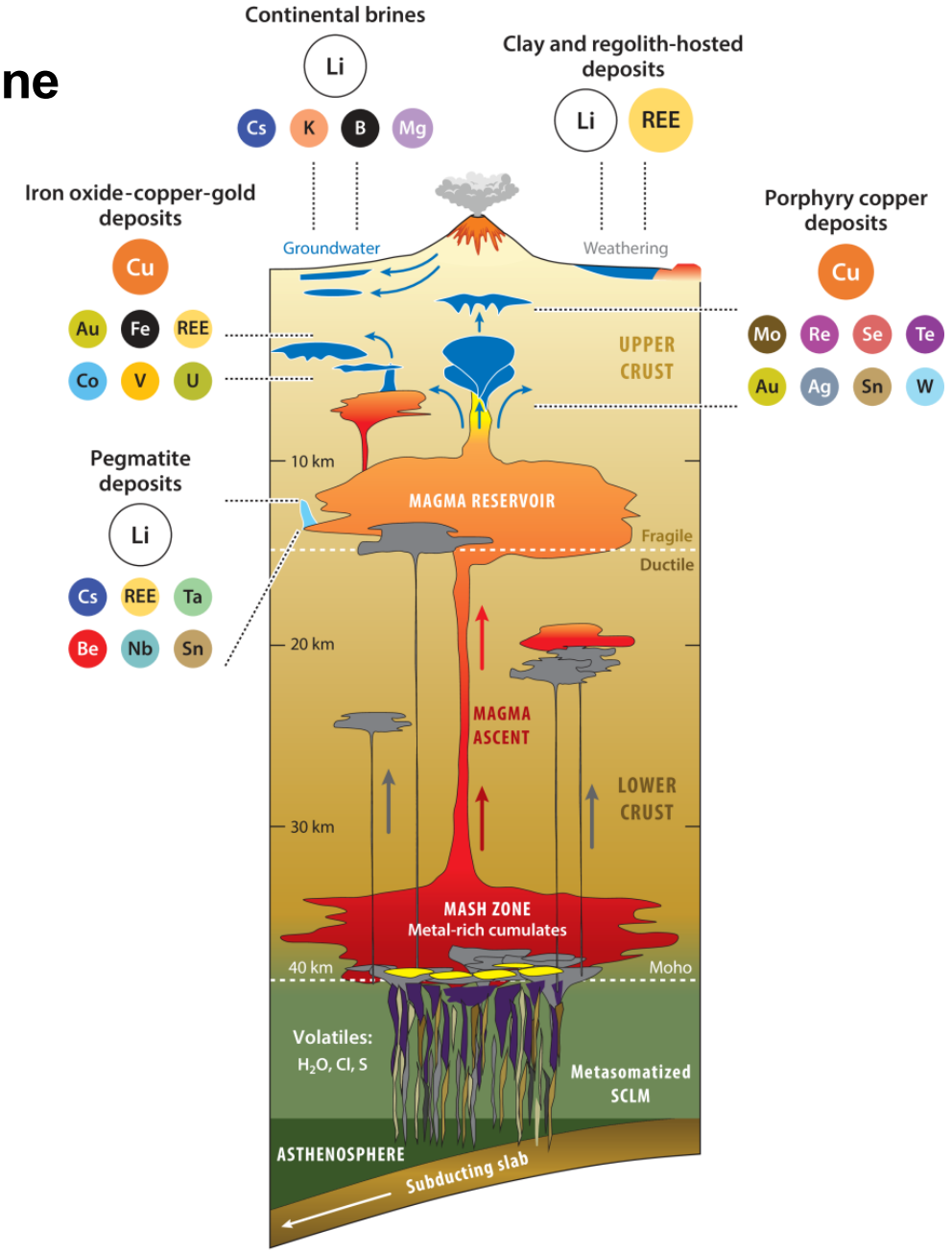


Source: https://bcgoldadventures.com



Geodynamic trends
Convergent margin/subduction zone

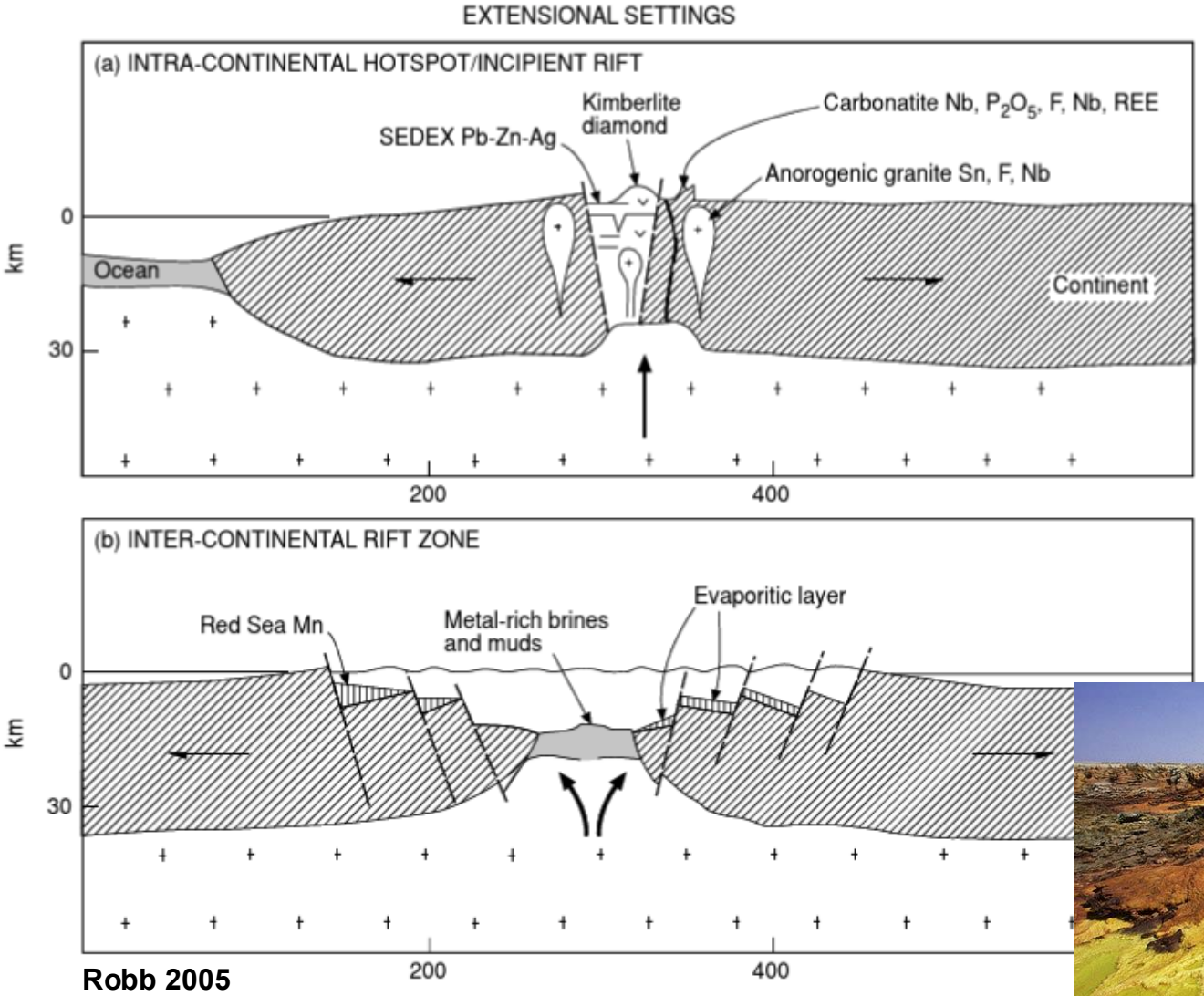
- **Processes:**
  - partial melting, intrusive bodies, metamorphism
  - Exsolution of hot magmatic-hydrothermal fluids, mixing with other crustal fluids, thermal circulation
  - Erosion of orogen lithologies
- **Deposits:**
  - Orogenic deposits (Au, base metals)
  - Crystallization and alteration in intrusions (e.g. U, Sn, W, Li)
  - Cu-Au-Mo porphyries
  - Placer
  - Sediment-hosted hydrothermal deposits (e.g. Cu, U, MVT)



Reich M, Simon AC. 2025
Annu. Rev. Earth Planet. Sci. 53:141–68

Reich 2025

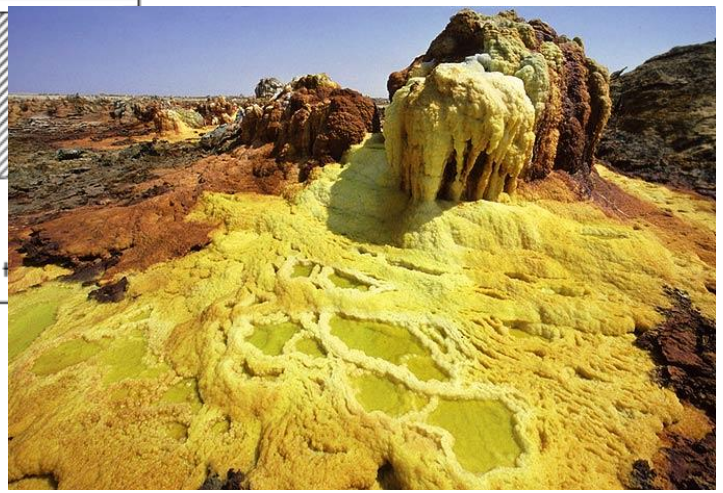
Geodynamic trends
Divergent margin



Robb 2005



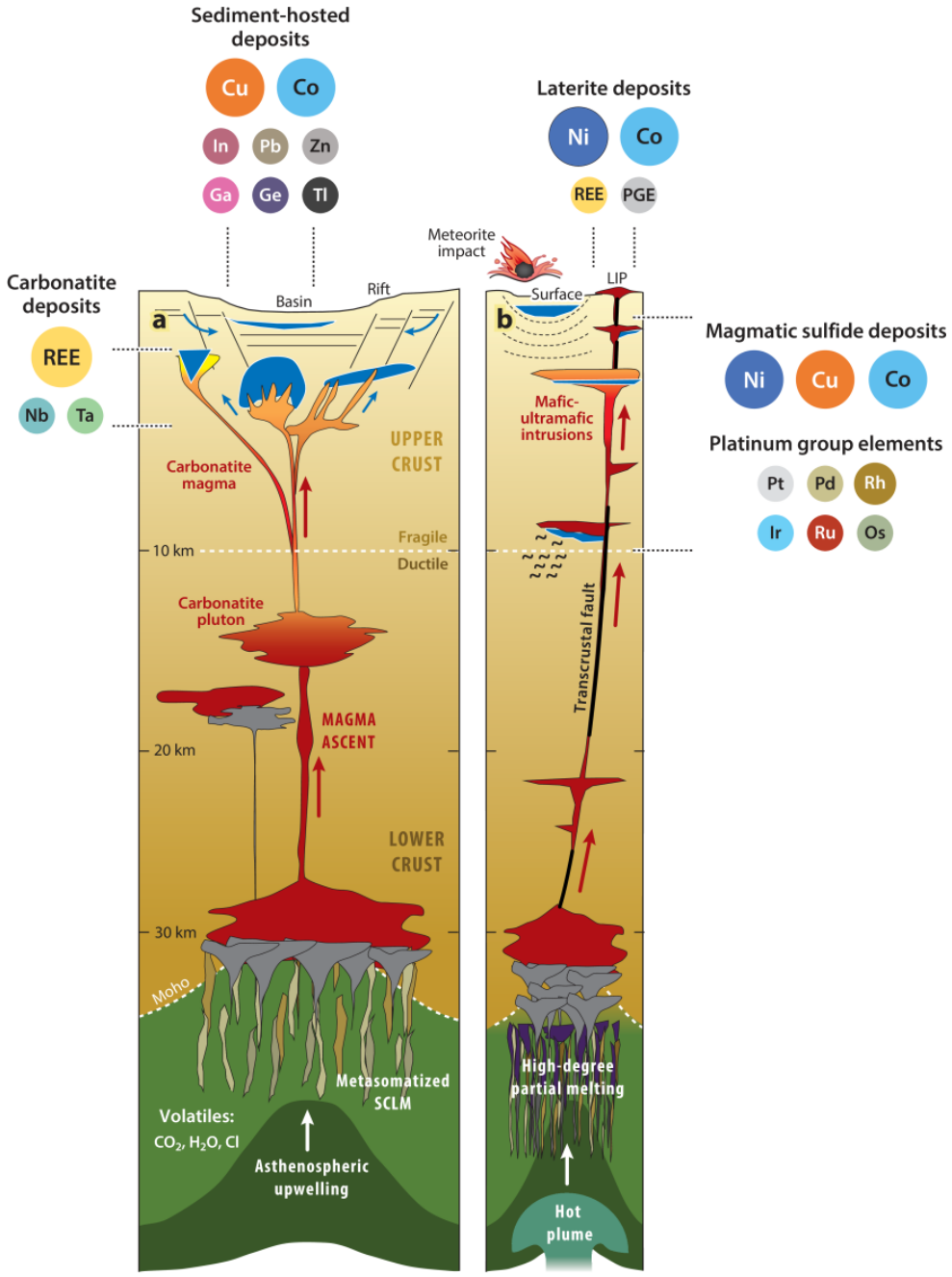
Source: Lucas Kavanagh/WHOI



Source: www.swisseduc.ch

Geodynamic trends
Divergent margin

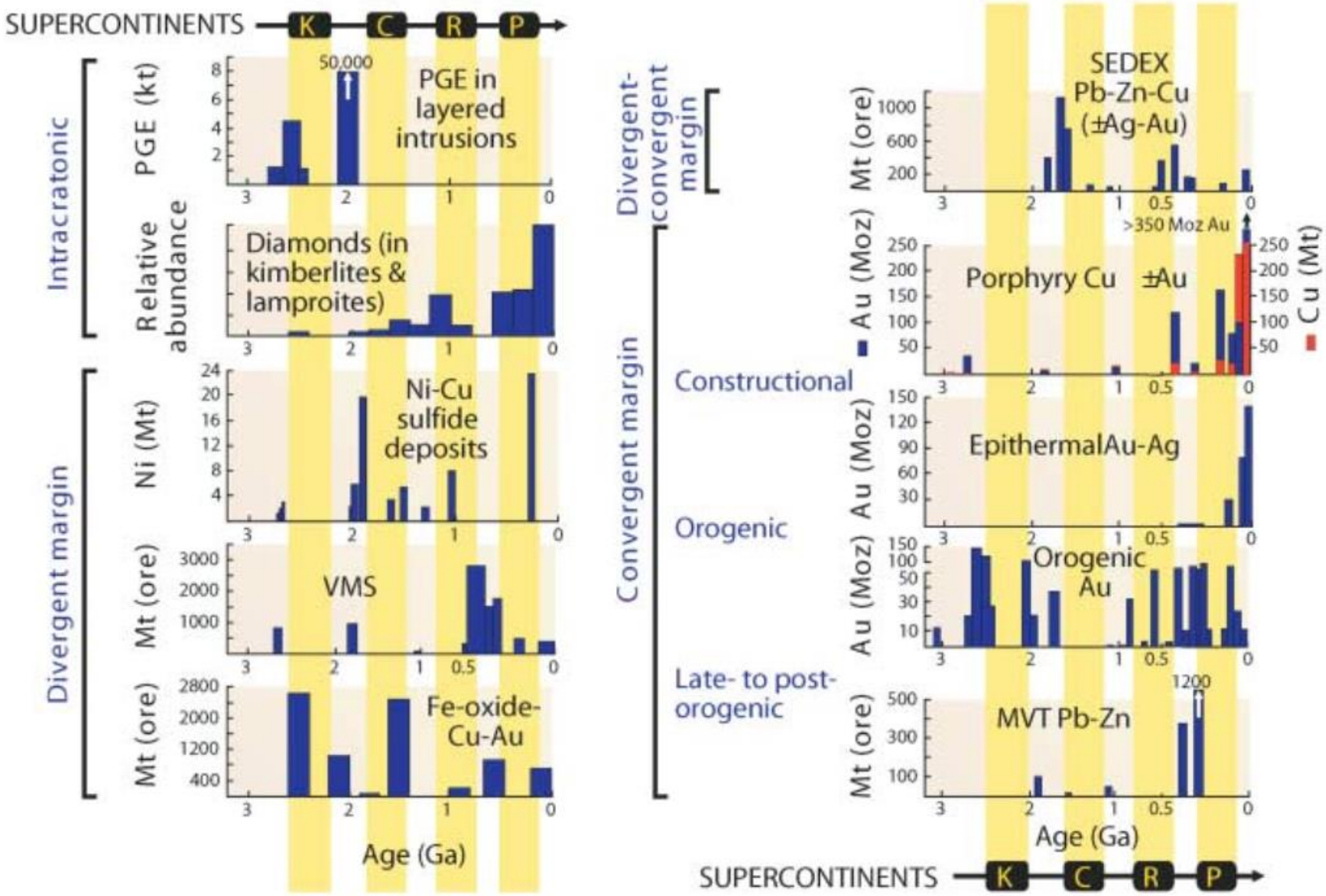
- **Processes:**
  - **Mantle plumes and partial melting**
  - **Exsolution of hot magmatic-hydrothermal fluids, mixing with other crustal fluids, thermal circulation**
  - **Fluid circulation through OC (’black smokers’)**
  - **Basement erosion**
  - **Sedimentary processes**
- **Deposits:**
  - **Ni-Cu-PGE sulfides in mafic/ultramafic intrusions**
  - **Sedex/VMS deposits (Pb-Zn, Cu)**
  - **Marine/continental deposits: BIF, hydrocarbons, evaporites**



Reich M, Simon AC. 2025
Annu. Rev. Earth Planet. Sci. 53:141–68

Reich 2025

Geodynamic trends
Crustal growth and ore deposits



Groves 2007

Global geological trends for commodity deposits
Key learning points week 11

- All commodity deposits represent suitable combinations of favorable geological processes, but similar processes created similar deposits through time. Hence, knowledge of these processes helps predict deposits.
- Resource deposits are geologically complex, and a detailed understanding of formation mechanisms is essential. In addition, the same commodity can be accumulated via different processes, and/or different processes may affect a single deposit over time.
- A useful first-order classification takes into account fluid types and origins: igneous, igneous-hydrothermal, hydrothermal, sedimentary and surficial processes
- The general relation between rock-forming and ore-forming processes means that an understanding of overarching controls (atmosphere, biosphere, geosphere, hydrosphere) is indispensable in predicting and studying commodity deposits.
- Economic geology draws on and integrates concepts from diverse disciplines: (geo)physics, geochemistry, igneous and metamorphic geology, volcanology, sedimentology, paleontology, landscape evolution…...